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Can the form of sexual selection explain patterns of static weapon allometry expressed by alternative mating morphotypes?

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Abstract

The regime of selection acting on a trait is expected to shape its static allometry. Few studies, however, have quantified the form of sexual selection acting on a trait in the wild to test whether the trait allometrically scales as predicted. Even fewer studies have tested these predictions using males expressing weapon polymorphism as part of their alternative mating strategies. Here, I use field data to test how sexual selection shapes scaling allometries of male weaponry in the Wellington tree wētā (*H. crassidens*), a male-trimorphic and harem-polygynous insect endemic to New Zealand. Contrary to the prediction that 10th instar males' large weaponry would scale hyperallometrically because it is under direct sexual selection, I found that 10th instar weaponry is not subject to direct sexual selection and scales hypoallometrically. Similarly, neither 8th nor 9th instar male weaponry experiences direct sexual selection, and their weaponry scales hyperallometrically and hypoallometrically, respectively. My study suggests that disentangling competing hypotheses for the evolution of scaling patterns of sexually selected traits must go beyond a simple viability-sexual selection dichotomy by also considering weapon function and the ecological context within which the weapon is used.

Keywords: allometry, polymorphism, reproductive strategies, sex, sexual selection

Introduction

Secondary sexual characters often display positive static allometries (e.g., Cothran & Jeyasingh, 2010; Eberhard & Gutierrez, 1991; Forsyth & Alcock, 1990; Gerstenhaber & Knapp, 2022; Green, 1992; McCullough et al., 2015; Painting & Holwell, 2013; Petrie, 1992; Simmons & Tomkins, 1996; Voje, 2016). Positive allometry of a secondary sex character is expected to evolve when the combined forces of viability and sexual selection provide a net fitness benefit to larger individuals producing relatively larger traits (Bonduriansky & Day, 2003; Kodric-Brown et al., 2006). If the secondary sexual character also serves a viability-related function, then survival will be maximized at a certain ratio of trait size to body size, and isometry or hypoallometry should prevail. Indeed, traits serving dual functions, such as the legs of males in several arthropods (Bonduriansky, 2007; Kelly, 2014b; Kelly, 2022), antennal length in male *Callosobruchus chinensis* beetles (Colgoni & Vamosi, 2006), wing size in Japanese beetles (*Popillia japonica*) (Kelly, 2022), and the forelimbs in several anuran amphibians (Schulte-Hostedde et al., 2011) are isometric or hypoallometric. Interestingly, however, some traits in insects that serve a dual function are hyperallometric. For example, the mandibles of male field crickets, which are used for feeding and fighting, tend to scale hyperallometrically (e.g., Judge & Bonanno, 2008; Kelly & L'Heureux, 2021; but see Bertram et al., 2021).

Another taxonomically widespread evolutionary outcome of intense sexual selection is the production of alternative phenotypes in which individuals within a sex (i.e., unconventional males) accrue mating success by adopting mating strategies that differ from those of conventional males (Gross, 1996; Oliveira et al., 2008; Shuster & Wade, 2003; West-Eberhard, 2003). Many species exhibiting alternative mating strategies have a conventional morph in which males defend or monopolize female mates using armaments, while the unconventional morph has males that either display a significantly reduced version of the trait or none at all. Conventional males are expected to experience strong, direct pre-copulatory sexual selection on their armaments because males having larger traits accrue greater mating success (e.g Brockmann, 2008). Therefore, the weaponry of these males might scale hyperallometrically if they are single-function (i.e., do not impose viability costs on their bearer). On the other hand, the armaments (if present) of unconventional males are likely to experience less intense direct sexual selection because the fitness of these males does not rely solely on weapon size. Consequently, sexual traits of unconventional males that are homologous to those in conventional males are more likely to be isometric or hypoallometric (i.e., negatively allometric). Empirical tests of the morph-specific scaling allometry of weaponry have heretofore been taxonomically limited (Kochensparger et al., 2024). Unfortunately, the form of sexual selection operating on the armaments

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in these taxa is unknown, so it is unclear how sexual selection might shape the static allometry of weaponry. For example, the forceps of male European earwigs is a dual-function trait involved in obtaining copulations (i.e., sexual function) and defending against predator attacks (i.e., survival function). Despite this trait imposing viability costs, conventional (i.e., macrolabic) male forceps scale hyperallometrically (e.g., Forslund, 2003; Simmons & Tomkins, 1996; Tomkins et al., 2005), contrary to prediction, while those of unconventional (i.e., brachlylabic) males scale isometrically (e.g., Forslund, 2003; Simmons & Tomkins, 1996; Tomkins et al., 2005; see also Kamimura, 2014), in line with prediction. Given that no study has identified the morph-specific forms of sexual selection operating on forceps in this species, it is not possible to infer how sexual selection shaped their static allometry.

How sexual selection produces one scaling pattern rather than another in dual-function traits is unclear. Our lack of predictive power could be due to a paucity of empirical examples with which to derive general patterns, which might, in turn, be a product of general neglect by empiricists of the allometric scaling patterns of dual-function traits. Only a handful of studies (e.g., Bertin & Fairbairn, 2007; Kelly, 2014b; Kelly, 2022; Lieshout & Elgar, 2009) have attempted to use a trait's pattern of static allometry to infer its sexual selection regime (or vice versa), with none strongly supporting the predicted process or outcome (Bonduriansky & Day, 2003). Species exhibiting alternative mating phenotypes within a sex present an excellent opportunity to test how a sexual selection regime shapes the static allometry of a trait, not only because the different morphs face different sexual selection pressures but also because these cases control for potentially confounding factors such as the shared genome and environment. The Wellington tree wētā, *Hemideina crassidens*, presents such an opportunity.

The Wellington wētā (*H. crassidens*) is a flightless, nocturnal orthopteran insect that expresses genetically trimorphic (non-reversible) alternative mating strategies through precocial maturation (Kelly, 2024). Males reach sexual maturity at either the 8th, 9th, or 10th instar (females also sexually mature at the 10th instar) (Kelly, 2024; Spencer, 1995). The three male morphs persist in stable proportions across time (Kelly & Adams, 2010), suggesting that morph frequency in this population is at or near equilibrium, and the strategies have equal fitness (Nason & Kelly, 2020). This wētā is a forest-dwelling species endemic to New Zealand in which both sexes seek daytime refuge from predators in tree cavities (Field & Sandlant, 2001; Kelly, 2006c). Because tree cavities can be limited in nature, several wētā can reside in them simultaneously, which creates an opportunity for harem formation (Kelly, 2006c, d; Kelly, 2008a; Wey & Kelly, 2018). Such female clumping, therefore, offers males a chance to enhance their reproductive success by monopolising access to them (*sensu* Emlen & Oring, 1977). Harem monopolisation favoured sexual selection for large mandibular weaponry in adult males and led to the evolution of male-biased sexual dimorphism in head size (Kelly, 2005, Supplementary Figure S1). Adult male head size positively covaries with maturation instar, and male reproductive behaviour is also morph-specific (Field & Deans, 2001; Kelly, 2005, 2024; Spencer, 1995). Indeed, 10th instar male *H. crassidens* use their enlarged mandibles in combat with rivals for control of female-occupied galleries (Field & Deans, 2001; Kelly, 2006a, c, d), and males with larger mandibles accrue greater harem success (Kelly, 2005; Kelly, 2008a; Nason & Kelly, 2020), probably because larger mandibles are superior weapons (Kelly, 2006a, 2014a). Precocially maturing 8th instar males possess smaller bodies and small weaponry and circumvent the defences of larger males to sneak copulations or search the foliage for undefended females. Males precocially maturing at the 9th instar are an intermediate jack-of-all-trades morph that can sneak or fight depending on the situation (Wilson & Kelly, 2019).

The exaggerated mandibles of male *H. crassidens* are a sexually selected trait that do not appear to impose significant viability costs via reduced immune function (Kelly, 2014a; Kelly & Jennions, 2009) but might do so through diminished foraging efficiency or reduced locomotory performance (but see Kelly, 2006b). Male morphs vary in the intensity of sexual selection operating on their mandibles, given that mandibular weaponry is more important to accruing larger harems in some morphs (i.e., 10th instar) than others (i.e., 8th instar) (Kelly, 2005; Kelly, 2008a; Nason & Kelly, 2020). Previous work on this species has shown that adult male weapon size (i.e., head length) in *H. crassidens* is hyperallometric when all three morphs are pooled and that female head size scales isometrically (Kelly, 2005). Here, I also predict that female head size will scale isometrically, but that morph-specific analysis will demonstrate that the weaponry of 10th instar males will be hyperallometric because they are under direct sexual selection. Alternatively, if this trait imposes viability costs, then it should scale isometrically. Predicting the scaling relationships of 8th and 9th-instar males is more challenging. Although the weaponry of 8th- and 9th-instar males is unlikely to be under direct sexual selection, since these morphs fight rivals less often for control of harems, they will likely still need some form of weaponry (e.g., for self-defence). Therefore, because the effectiveness of their weaponry might be maximized at a particular weapon size-to-body size ratio rather than the largest weapon size possible, their weaponry should scale isometrically. However, even if their mandibles are not as exaggerated as those of 10th-instar males, they should still impose viability costs, which would also favour isometry. This is the first study, to my knowledge, that uses field data to empirically test whether the form of sexual selection acting on the weaponry of alternative mating morphotypes is reflected in their pattern of static allometry.

Methods

Study animals

I collated and used data collected in previous field studies between 2001–2017 on Maud Island (Kelly, 2005, 2008b; Kelly, 2008a; Kelly & Jennions, 2009; Nason & Kelly, 2020) in which the femur and head lengths of males and females were measured. Femur length is a proxy measure of body size. Head length can be considered a proxy measure of weapon size because it strongly correlates with mandible length (Kelly, 2005), the trait that physically engages with opponents in combat. However, the entire head could also be functionally considered a weapon, as the head capsule contains the jaw muscles that operate the mandibles (O'Brien & Field, 2001). Morphs with longer mandibles have larger

Table 1. Summary statistics for the head and femur lengths of female and three male alternative mating morphotypes in H. crassidens.

Sex/morph	Head length			Femur length		N
	x ± SD	CV	CV'	x ± SD	CV	
female	13.04 ± 0.90	6.87	4.11	20.87 ± 1.33	6.35	472
8	16.30 ± 1.38	8.49	5.59	16.75 ± 0.82	4.92	399
9	20.81 ± 1.31	6.29	5.21	18.48 ± 0.75	4.08	263
10	26.75 ± 1.68	6.27	4.80	20.48 ± 0.88	4.31	242

CV represents phenotypic variance and CV' represents the phenotypic variance adjusted for body size (i.e., femur length).

head capsules and thus a stronger bite force (Kelly, 2014a). In all years, both traits were measured to the nearest 0.05 mm using digital callipers (Mitutoyo Digimatic). As described in previous studies (Kelly, 2005, 2024; Kelly & Adams, 2010), n = 1134 male *H. crassidens* were assigned a morphotype by using a trimodal mixture model (Prates et al., 2013). I retained individuals that could be assigned with at least a 95% probability of correct assignment. Females mature at the 10^{th} instar and express a single morphotype having a normal distribution (Kelly, 2005; Kelly & Adams, 2010). Final per-year sample sizes for females and each male morph are given in Table S1.

Intra- and inter-sexual morphology

Statistical analyses were performed within the R v. 4.2.1 statistical environment (R Core Team, 2024). I tested sex and morph differences in head length and femur length using separate general linear models (GLMs). Each model had sex/morph as the explanatory variable and the appropriate response variable. I conducted planned contrasts to test differences between specific groups based on previous work (Kelly, 2005; Kelly & Adams, 2010). I assessed phenotypic variation in head and femur length for females and male morphs by calculating coefficients of variation (CV_p) as the standard deviation (σ) divided by the mean (μ). Differences between females and male morphs in CV_p were calculated using the R package *cvequality* (Marwick & Krishnamoorthy, 2021). I also calculated the coefficient of variation that head length would have if body size (femur length) were held constant: $CV' = CV(\text{head length}) \times (1 - r^2)^{0.5}$, where r is the Pearson correlation coefficient between head length and femur length (Eberhard et al., 1998).

Static allometry of weaponry

Whether a statistical method based on ordinary least squares (OLS) regression or geometric mean (e.g., major axis regression) regression should be used to estimate allometric slopes is a topic of ongoing debate (Al-Wathiqui & Rodrı´guez, 2011; Eberhard et al., 1999; Hansen & Bartoszek, 2012; Kilmer & Rodrı´guez, 2017; Voje, 2016). I use an extension of OLS, linear mixed-effects models, to calculate slopes separately for females and all three male morphs. Each of the four models included head length as the response variable, femur length as the independent variable, and sample year as the random effect (see Supplementary Table S1 for sample sizes). All traits were log10-transformed before analysis. The R package *emmeans* (Lenth et al., 2019) was used to calculate 95% confidence intervals for each slope. Slopes having intervals including 1.0 indicate isometry.

Sexual selection analysis

I used generalized linear models (GLMs) to calculate linear (β_i) and nonlinear (γ_{ii}) sexual selection gradients and their corresponding 95% confidence intervals by using the method of Morrissey & Goudie (2022). Generalized linear models were required because my fitness measure (mean harem size) produced non-normal residual distributions due to being real numbers, including zero. I calculated mean harem size per male for males contributing multiple records to the dataset. Because there was evidence of overdispersion, the GLMs were fit with a negative binomial family of errors using a log link function (Morrissey & Goudie, 2022). Linear (β_i) selection is the partial regression coefficient of a trait and is indicative of a partial change in the phenotypic mean. Non-linear (γ_{ii}) selection is the partial regression of squared terms and is indicative of a partial change in the variance of a trait. Correlational selection (γ_{ij}) is the partial regression coefficient of cross-product terms and is indicative of a change in the covariance between two traits (Brodie et al., 1995). My GLMs included mean harem success as the fitness variable, head and femur length as covariates (β_i), as well as their squared (γ_{ii}) and cross-product (γ_{ij}) terms. Separate models were run for each morphotype. Phenotypic traits were standardized (mean = 0, standard deviation = 1.0) prior to analysis. GLM parameter estimates were transformed into selection gradient estimates following Morrissey & Goudie (2022). Because data were collected over several years (Supplementary Table S2), I used negative binomial models having mean harem success as a response variable, head and femur lengths as covariates, the fixed factor year, and their interactions to test whether selection gradients varied across years. Separate models were run for each morphotype. Statistical tests using the *R* function *Anova()* suggested that year did not have significant interactive effects with any trait on gradient estimates in any of the three models (Supplementary Tables S3-S5); therefore, it was deleted. The relationship between fitness and phenotype was visualised using the *fitness.landscape* function in the *GSG* R package (Morrissey & Sakrejda, 2013).

Results

Intra- and inter-sexual phenotypic trait measures

Head length differs significantly among females and the three male morphotypes (one-way ANOVA: F = 6724.14, df = 3,1372, $p < 0.001$) with females having a smaller head size than 8^{th} instar males (Estimate ± SE = −6.19 ± 0.06, t = −112.05, $p < 0.001$), 8^{th} instar males having a smaller head size than 9^{th} instar males (Estimate ± SE = −9.11 ± 0.07, t = −126.40, $p < 0.001$), and 9^{th} instar males having a smaller head size than 10^{th} instar males (Estimate ± SE =

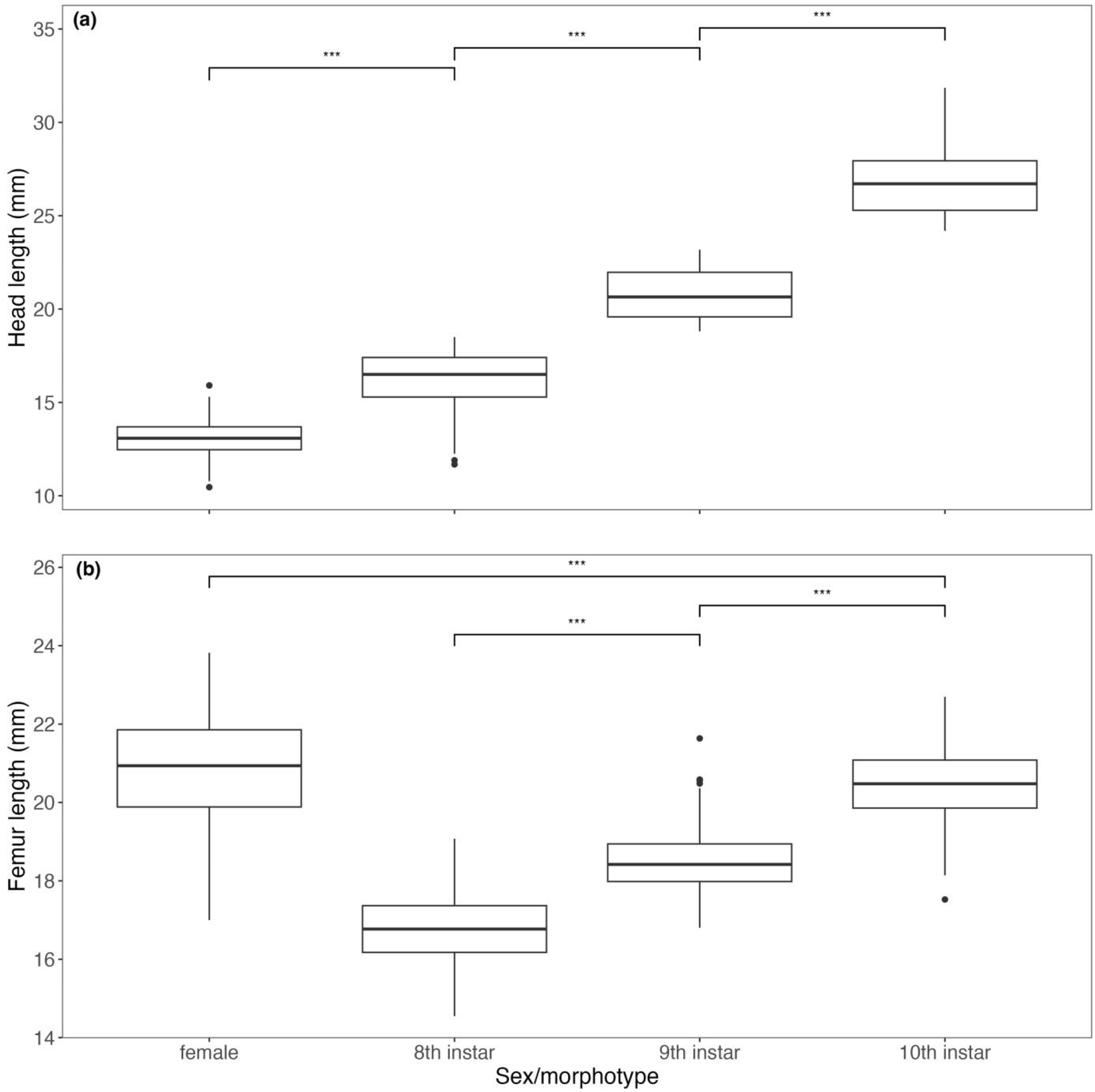

Figure 1. Boxplots showing sex/morphotype differences in (a) head length (mm) and (b) femur length (mm). The box represents the lower (25%) and upper (75%) quartiles, the solid dark horizontal line is the median, and the whiskers indicate 1.5 times the size of the hinge, i.e., the 75% minus 25% quartiles. See Table 1 for sample sizes, * $P < 0.0001$.

-7.53 ± 0.07, $t = -109.60$, $p < 0.001$; Table 1; Figure 1a). Phenotypic variance (CV_p) of head length in 8th instar males was greater than in females ($p < 0.001$), which in turn was greater than in 9th instar males ($p < 0.001$); 9th and 10th instar males did not differ statistically in their degree of head length variance ($p > 0.9$) (Table 1). When adjusting for body size (CV′), females have the smallest CV overall, while 8th instar males still have the largest CV among the three male morphs (Table 1).

Females have significantly longer femora than 10th instar males (Estimate ± SE = 1.72 ± 0.04, $t = 39.22$, $p < 0.001$), 10th instar males have significantly longer femora than 9th instar males (Estimate ± SE = 3.06 ± 0.06, $t = 53.34$, $p < 0.001$), and 9th instar males have significantly longer femora than 8th instar males (Estimate ± SE = 2.39 ± 0.05, $t = 51.83$, $p < 0.001$; Table 1; Figure 1b). Phenotypic variance (CV_p) of femur length in females was greater than in 8th instar males ($p < 0.001$), which in turn was greater than in 9th instar males ($p < 0.001$); 9th and 10th instar males did not differ statistically in their degree of head length variance ($p > 0.9$; Table 1).

Head length static allometry

Female head length scaled hypoallometrically (slope = 0.87, 95% CI = [0.81, 0.93], Figure 2; see Table 1 for sample sizes).

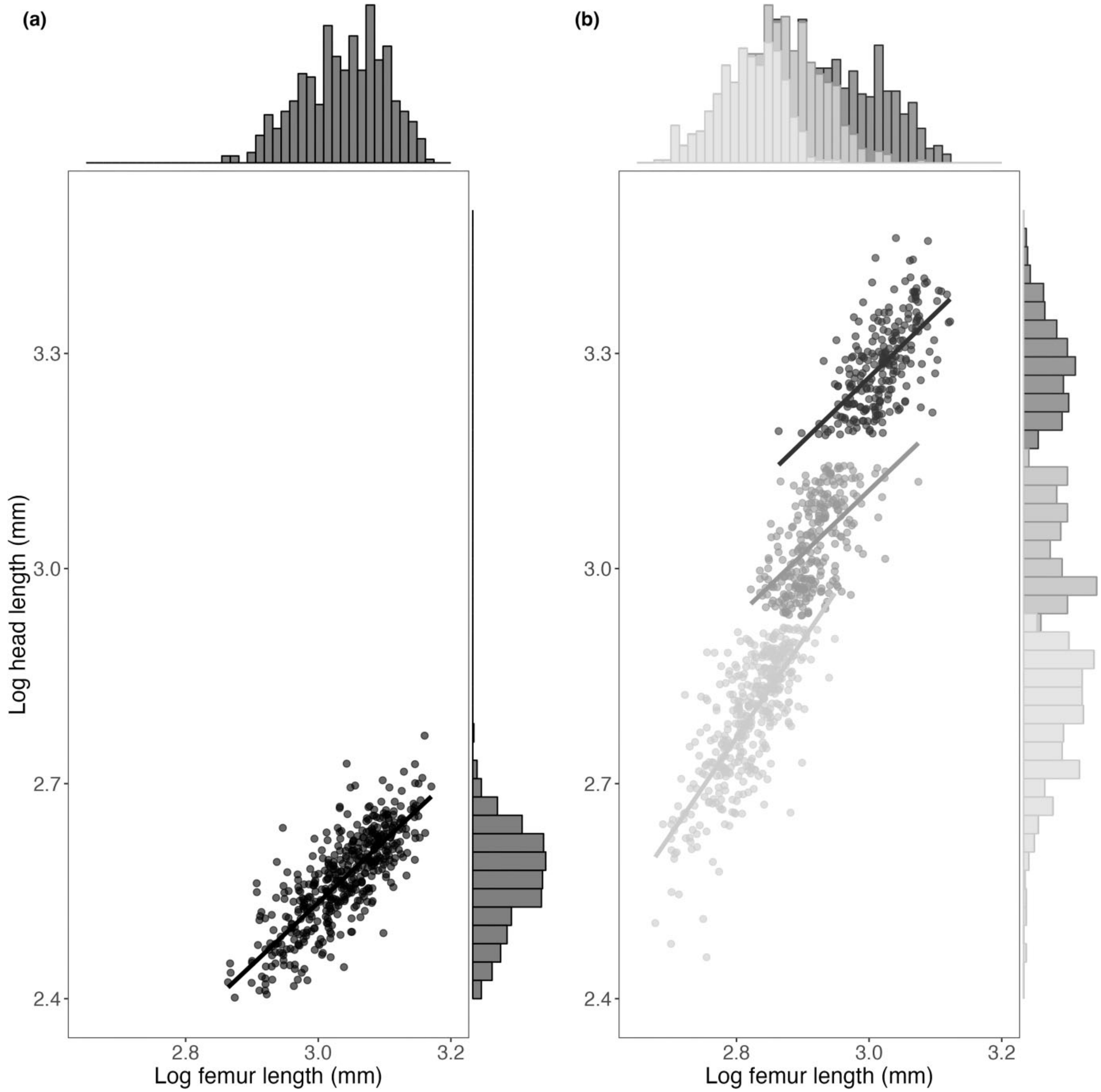

Figure 2. Allometric relationships between head length (log10 transformed) and femur length (log10 transformed) in (a) female and (b) male H. crassidens. Scaling relationships are shown in (b) for each morphotype separately: 8^{th} instar males (light grey); 9^{th} instar males (dark grey), and 10^{th} instar males (black). Trait distributions are shown on the top and right sides in both plots. Allometric slopes were obtained using ordinary least squares regression. See Table Table 1 for sample sizes.

The static allometry of head length for pooled male morphs was significantly positive (slope = 2.19, 95% CI = [2.14, 2.25]). Examination of static allometric scaling for each morphotype separately revealed that head length scales hyperallometrically in 8^{th} instar males (slope = 1.36, 95% CI = [1.240, 1.479]; Figure 2; Supplementary Table S6). The slopes for 9^{th} (slope = 0.89, 95% CI = [0.730, 1.045]) and 10^{th} (slope = 0.89, 95% CI = [0.751, 1.037]) instar males suggest isometry since the 95% CIs include 1.0. However, these slopes are more in line with hypoallometry given that they are ~ 0.89 with upper limits very close to 1.0 (Figure 2; see Table S6 for complete model outputs).

Selection analysis

Multivariable selection analysis supported neither direct linear nor non-linear sexual selection on head length in any morph (Table 2; Supplementary Figures S2 and S3). Femur length, however, was under strong direct negative selection in 8^{th}-instar males (Table 2; Supplementary Figure S3). I found no evidence for bivariate nonlinear selection in any morph (8^{th}: $\gamma_{ij} = -0.43 \pm 0.35$, $p > 0.05$; 9^{th}: $\gamma_{ij} = -0.41 \pm 0.49$, $p > 0.05$; 10^{th} instar: $\gamma_{ij} = -0.61 \pm 0.46$, $p > 0.05$; Table 2).

Table 2. Linear (β_i) and nonlinear (γ_{ii}) sexual selection gradients (±SE) for male weapon (head length) and body size (femur length) in Hemideina crassidens.

Trait	8th instar		9th instar		10th instar	
	$\beta_i \pm$ SE	$\gamma_{ii} \pm$ SE	$\beta_i \pm$ SE	$\gamma_{ii} \pm$ SE	$\beta_i \pm$ SE	$\gamma_{ii} \pm$ SE
head length	0.23 ± 0.20	0.47 ± 0.44	−0.41 ± 0.33	0.81 ± 1.12	0.22 ± 0.21	0.48 ± 0.42
femur length	−0.49 ± 0.19*	0.32 ± 0.44	0.20 ± 0.21	0.12 ± 0.29	−0.15 ± 0.22	0.68 ± 0.53

Gradients are calculated from generalized linear model regression analyses with a log-link function and standardized traits (Morrissey & Goudie, 2022). Sample sizes: 8th instar = 252; 9th instar = 160; 10th instar = 141. * $P < 0.05$.

Discussion

Head size in male and female Wellington tree wētā (*H. crassidens*) faces different selection pressures, with viability selection acting on female head size and sexual selection being the main evolutionary force shaping resource allocation to male heads. In this study, I first tested the hypothesis that these different selection pressures should result in a sex difference in the static allometric scaling of head size. I then tested the hypothesis that scaling patterns differ among alternative mating morphotypes because the role that weaponry plays in reproductive success accrual differs among morphs. Finally, I assessed whether the observed morph-specific scaling patterns of male mandibular weaponry align with the observed form of sexual selection operating on this trait. I found that female head length scales hypoallometrically (negative static allometry) in *H. crassidens*, whereas the common slope (all three morphs pooled) for males is hyperallometric. The current finding of male hyperallometry using OLS confirms the finding of a previous study that used major axis regression to calculate the allometric slope (Kelly, 2005). These two findings agree because the slopes are very steep in both cases. In contrast, because the model II regression female slope in Kelly (2005) was borderline isometric and because model II regression tends to overestimate slopes (Al-Wathiqui & Rodrı´guez, 2011), the current study, using the more conservative OLS approach, suggests that female head size scales hypoallometrically. My current morph-specific analysis shows that females have a lower allometric slope than 8th instar males, but the same scaling pattern as 9th and 10th instar morphs. The hypoallometry expressed by female *H. crassidens* is common in animals (Voje, 2016) and is likely driven by a combination of factors. For example, head size likely reaches a naturally selected optimum (sensu Lande, 1980) via foraging efficiency, so a constant linear increase in head size could impose significant viability costs. However, female *H. crassidens* should accrue greater fitness benefits via increased fecundity by increasing body size (Kelly & Jennions, 2009). Consequently, female viability increases with body size but decreases with head size (see Bonduriansky & Day, 2003).

Eberhard et al. (2018) argue that weapon allometry should be positive, particularly if the trait also functions as a threat-signalling device or if exaggerated weaponry provides a biomechanical advantage, such as allowing the bearer to contact a smaller rival at a greater distance. Contrary to this prediction, I found that the weaponry of 10th (and 9th) instar male *H. crassidens* do not scale positively, possibly because their role in threat signalling is limited, given that male-male interactions occur at night and often in complete darkness. Additionally, positive allometry could be disfavoured by selection because greatly exaggerated weaponry could limit a male's ability to enter a tree cavity for mates or refuge, or because male-male combat in *H. crassidens* often occurs within the restricted confines of a tree cavity, which would limit any possible biomechanical advantage conferred by positively allometric mandibles. Rather, my analyses show that weapon allometry is, contrary to prediction, hyperallometric in 8th instar (unconventional) males only, while 9th (unconventional) and 10th instar (conventional) male weaponry scales hypoallometrically. The hypoallometry of 9th and 10th instar male weaponry is perplexing since males will grab the hind legs of female mates (Kelly, 2008c) and male rivals (Kelly, 2006a) and pull them from galleries for mating or fighting, respectively, which should favour longer mandibles. Thus, I would expect 9th and 10th-instar males to at least have weapons that scale isometrically. Instead, perhaps positive and linear allometry are selected against because the pinching or grasping efficiency is weaker in longer structures (Eberhard et al., 2018). Therefore, the bite force of 10th (and possibly 9th) instar males might be a decreasing function of weapon size and an increasing function of body size, a relationship that would produce hypoallometric scaling (see Bonduriansky & Day, 2003).

That an unconventional morph (8th instars) has a steeper allometry than the conventional morph resembles the pattern exhibited by the alternative morphs in the harvestman, *Phareicranaus manauarais*, where minor (unconventional) males have positively allometric tusks, and those of major (conventional) males are isometric (Palaoro et al., 2022). My findings, however, contradict those in another hemimetabolous insect, *Forficula auricularia*, where macrolabic (conventional) male forceps length scales positively allometrically (Forslund, 2003; Simmons & Tomkins, 1996; Tomkins et al., 2005) while those of brachylabic (unconventional) males scale isometrically (Simmons & Tomkins, 1996; Tomkins et al., 2005; but see Forslund, 2003). Caution must be exercised in drawing conclusions from these comparisons, however, as these studies used a mix of model I and model II regression techniques to calculate allometric slopes. I could not find any examples of hemimetabolous insects with unconventional males exhibiting positively allometric weaponry and conventional males with a different scaling pattern, as observed in the current study. This pattern, however, is common in holometabolous insects (McCullough et al., 2015) and is generally attributed to resource depletion; major males invest in a larger body size and therefore do not have the extra nutritional resources available to allocate to a proportionately larger weapon (Nijhout & Wheeler, 1996). However, the weapon hypoallometry of the 10th instar *H. crassidens* males is unlikely to be a consequence of resource depletion because, as a hemimetabolous insect, males acquire resources for growth at every stage of development, including before the moult to adulthood. Tomkins & Simmons (1996) suggested that the positive allometry of macrolabic (conventional) male earwigs (*F. auricularia*) was proximally possible because morph expression in this species is condition-dependent. In other words, males in better condition not only express the macrolabic strategy, but larger and better-conditioned macrolabic males can allocate proportionally more resources to forceps length (see also Bonduriansky, 2007). This proximal argument does not hold, at least in part, based on my observations, because morph expression in *H. crassidens* is genetically polymorphic rather than a conditional strategy (Kelly, 2024). That said, male body and weapon size within a morph could be condition-dependent, with individuals exhibiting larger trait values at one instar having better body condition in the previous instar. Although we do not know the condition-dependence of trait expression in *H. crassidens*, we can infer its role based on phenotypic variation. Sexually selected traits are expected to be condition-dependent because they are costly to produce, and if condition is dependent on many underlying loci, then genetic and phenotypic variance should be higher for sexually selected traits than for non-sexual traits (Pomiankowski & Møller, 1995; Rowe & Houle, 1996). Although head size in *H. crassidens* is a sexually selected trait, I found that it was significantly more phenotypically variable (as measured by the phenotypic coefficient of variation CVp) in 8th instar males than in either females or the other male morphs, suggesting that it might be more condition-dependent in 8th instar males than in the other male morphs. I also found that a non-sexual trait, femur length, was significantly more variable in 8th-instar males and females than in 9th and 10th-instar males. These analyses, therefore, suggest that perhaps the 8th instar male phenotype is more condition-dependent than that of the other two morphs and that female body size might also be condition-dependent.

My sexual selection analyses offer little evidence that male head size is strongly shaped by sexual selection within any morph, as males with comparatively larger weapons in each morph do not have larger harems. Instead, I argue that sexual selection more effectively operates through sexual competition among morphs, with 10th instar males gaining higher mating success by defeating other morphs for access to harems, while 8th and 9th instar males derive most of their fitness through means other than harem defence (Kelly, 2005; Nason & Kelly, 2020). Certainly, it is probably quite difficult for there to be strong direct sexual selection on weapon size within an *H. crassidens* morph because social groups are spatially and temporally fluid (Kelly, 2006c; Kelly, 2006b), which means that it is unlikely that males with the largest weaponry (i.e., greater resource-holding potential, RHP) will consistently find and monopolise the largest groups of females each night (Kelly, 2006d; Kelly, 2006b, 2008a; Wey & Kelly, 2018; see also McDonald et al., 2013). Indeed, male *H. crassidens* with greater RHP were unable to consistently monopolise access to artificial galleries housing the most females in relatively small experimental cages (Kelly, 2006a), and males having the greatest RHP did not always control the galleries hosting the most females within small forest patches (Kelly, 2006d). Second, hard selection (sensu Wallace, 1975) likely determines the pool of individuals available to compete for certain harem-occupied galleries. In other words, although all males will reside with a harem if possible, not all males can gain access to every harem. Access to a gallery is determined by the size of the gallery's entrance hole. Females have significantly narrower heads than males, meaning that they can enter cavities via entrances that might preclude larger-headed males. Therefore, perhaps only 8th instar and smaller 9th instar males can gain access to most female-occupied galleries, whereas larger 9th instars and 10th instars cannot. It is even possible that smaller 10th instar males can gain access to harems while larger males cannot, which might explain why even this fighter-guarder morph lacks significant directional sexual selection on weapon size. Finally, the fact that 8th and 9th-instar males can reside with large harems might be due to their relative abundance in the population. These morphs, particularly 8th instar males, are significantly more abundant on Maud Island than 10th instar males (Kelly & Adams, 2010). Therefore, smaller instars, which wander in search of mates, are perhaps more likely to find an undefended harem. Combined, these factors lead to 8th and 9th-instar males occasionally residing with large harems that would otherwise have been controlled by a 10th-instar male had he found it. These random variations notwithstanding, harem success is broadly correlated with morphotype, with 10th-instar males having the greatest success, on average, and 8th-instar males achieving the least (Nason & Kelly, 2020). Despite the inequality of average harem sizes among morphs, the ability of 8th and 9th-instar males to at least occasionally reside with harems likely contributes to their having average fitnesses equal to 10th-instar males, which ultimately maintains the alternative morphs in this population (Nason & Kelly, 2020).

The absolute magnitude of linear (0.23, 0.41, and 0.22) and quadratic (0.47, 0.81, and 0.48) selection gradients reported here for head length are much larger than the median values of 0.16 and 0.10, respectively, that are reported in the literature (Kingsolver et al., 2001). My selection analyses quantified only the magnitude of sexual selection operating on male weapon size through harem size. Total sexual selection on weaponry could be greater than what was measured here if larger weaponry is also favoured via routes other than fighting. For example, perhaps sexual selection also favours larger mandibles in each morph because they permit males to more easily extract females from galleries for mating or restrain them during copulation (Kelly, 2008c). Furthermore, it is important to recognise that although fighting for and controlling mating access to harems comprises a major part of a 10th instar male's fitness-acquisition repertoire, it is likely a minor part of an 8th or 9th instar male's repertoire. For example, 8th and 9th-instar males search for and copulate with females outside of galleries (e.g., on tree trunks and foliage) and sneak matings with some members of a harem guarded by a 10th-instar male, and despite having lower mating success, they will maximise their per female fertilisation success by investing more in ejaculates than 10th-instar males (Kelly, 2008b; Nason & Kelly, 2020). My selection analyses, therefore, potentially capture only a portion of sexual selection operating on male head size and only one facet of sexual selection operating on males in general.

Weapon size in *H. crassidens* was shown by Kelly (2008a) to experience direct sexual selection via harem size under specific ecological conditions (i.e., environments having large galleries), and this selection regime appeared to functionally explain the hyperallometric scaling of weaponry in this species (Kelly, 2005). That analysis of static allometry, however, pooled all males (i.e., all morphs combined). In contrast, the current morph-specific analyses revealed that only 8th-instar males have hyperallometric weaponry. Powell et al. (2020) similarly found that pooled samples of male harvestmen (*Forsteropsalis pureora*) exhibit positive weapon allometry, but morph-specific analyses exhibit a mix of hypoallometric and isometric patterns. My morph-specific analyses, therefore, raise the question as to whether sexual selection, or some other process, shapes weapon allometry. The allometries observed in 8th and 9th-instar male *H. crassidens* might have emerged before the evolution of precocial sexual maturation and alternative mating strategies and thus might not be shaped by sexual selection operating on each morph in isolation. Perhaps, prior to the evolution of the three alternative strategies, a mutation arose that reprogrammed the allometric scaling of head size in 8th instar males (cf. Nijhout & Wheeler, 1996; see also Tomkins et al., 2005) that facilitated the production of larger weaponry in 10th instar males. This reprogramming could have been favoured by sexual selection because larger (and perhaps better conditioned) immature 8th-instar males having proportionately larger heads would later mature at the 10th instar with significantly larger weaponry. Those males with this mutation would likely outcompete those without it (i.e., those scaling isometrically), as the latter would mature at the 10th instar with significantly smaller weaponry. If this ontogenetic scaling pattern is canalized then it could mean that although it ensures that 10th instar males express larger weaponry, it might also constrain the ability of 8th and 9th instar males to independently alter the static scaling of their weaponry in response to sexual and viability selection. This hypothesis assumes that the heads of immature 8th and 9th-instar males scale identically to their sexually mature counterparts, which remains to be tested.

Alternatively, perhaps the morph-specific allometry of weaponry is evolutionarily unconstrained, with the observed allometries of each morph being shaped by sexual and viability selection. If so, then the hyperallometry of 8th instar weaponry is predicted to be a consequence of direct sexual selection such that larger males accrue greater fitness benefits by investing proportionally more in a sexually selected trait, whereas smaller males do not gain such benefits (example 5 in Bonduriansky & Day, 2003; Green, 1992; Petrie, 1992; see also Bonduriansky, 2007; Eberhard et al., 2018; Shingleton & Frankino, 2013). This, however, does appear to be the case as I found no evidence of direct sexual selection operating on 8th instar weaponry, at least when the fitness measure is harem size. Similarly, hypoallometric scaling of 9th instar weaponry requires direct sexual selection on weapon size, with viability increasing with body size but decreasing with weapon size (see Bonduriansky & Day, 2003). That I found no evidence of direct sexual selection for larger harems operating on 9th instar weapon size does not support this explanation. If sexual selection has any role in shaping the allometric scaling of weaponry in 8th and 9th-instar males, it does not appear to be through fighting for larger harems.

The weaponry of 10th instar males scales hypoallometrically. This form of scaling is usually a product of direct sexual selection on trait size, accompanied by concomitantly increasing viability costs on the weapon (Bonduriansky & Day, 2003). Despite there being weak direct sexual selection operating on 10th instar weaponry, it is still possible that viability selection favours larger males with less exaggerated weaponry. Examples of viability costs acting against 10th instar weaponry include reduced locomotory performance, diminished foraging efficiency (e.g Oufiero & Garland, 2007; Shingleton & Frankino, 2013), and limited access to diurnal refuges with relatively small entrance holes (males that are denied access to diurnal refuges face significant survival costs via increased predation). Hypoallometry could be favoured in lieu of hyperallometry if the latter experiences weaker bite forces. Direct tests of the costs of exaggerated weaponry in this and other species are necessary.

In conclusion, the current study is similar to others showing that the observed selection regime does not align with the predicted scaling pattern of a sexually selected trait (Bertin & Fairbairn, 2007; Kelly, 2022; but see Kelly, 2014b). Rather than males with larger weaponry within each morph controlling larger harems, I suggest that sexual selection on weapon size is likely operating more strongly at the level of morphotype, such that 10th instar males defeat the other morphs in combat and thus acquire larger harems, on average, than the other morphs. My study suggests that a comprehensive understanding of how sexual selection shapes weapon allometry within each alternative morphotype in *H. crassidens*, and other species, will require more than quantifying how sexual and viability selection relate to weapon and body size. Rather, complete explanations will require knowing how weapons are specifically used by males to accrue fitness and how that function, and ultimately the weapon's morphology, is shaped by the species' ecology. In *H. crassidens*, for example, ecology will play a greater role in shaping weapon morphology in 10th instar males because, despite sexual selection favouring larger weaponry in this morph, weapon size will ultimately be dictated by access to harem-holding tree cavities. 8th and 9th-instar males are unlikely to face such ecological constraints. Further research is needed to understand how sexual and viability selection, together with weapon function and ecology, interact to influence allometric scaling patterns of weaponry.

Supplementary material

Supplementary material is available online at *Evolution*.

Data availability

Data are available through the OSF repository at 10.17605/OSF.IO/MFJ8E.

Author contributions

C.D.K. collected and analyzed the data and wrote the manuscript.

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Conflict of interest

I declare no conflict of interest.

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